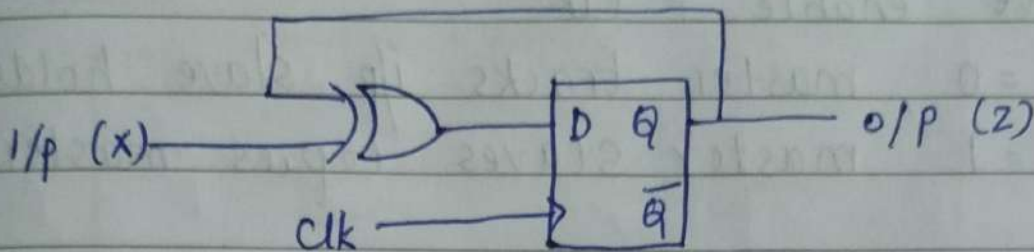


Assignment 4

Sequential Assignment

1) Explain the behaviour of the given circuit.



Present (Q)	Input X	Q _{next}
0	0	0
0	1	1
1	0	1
1	1	0

$$D = X \oplus Q$$

$$Q_{next} = D$$

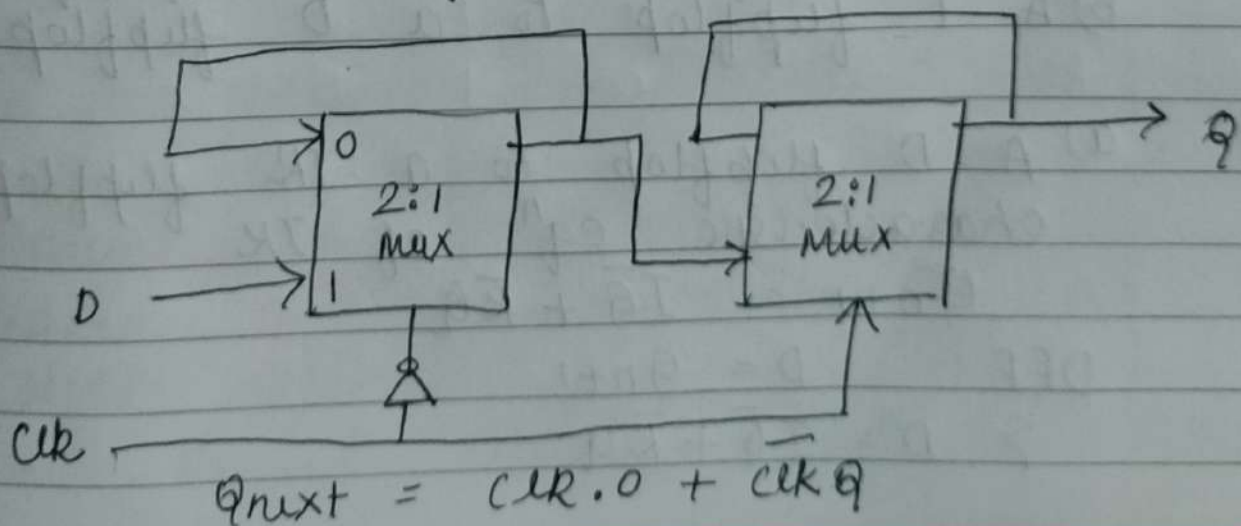
$$Q_{next} = X \oplus Z$$

$$i) x=0, Q_{next}=Q$$

$$ii) x=1, Q_{next}=\bar{Q}$$

$\therefore$ It is a T flipflop

2) Design a posedge triggered D flipflop using 2:1 Muxes only



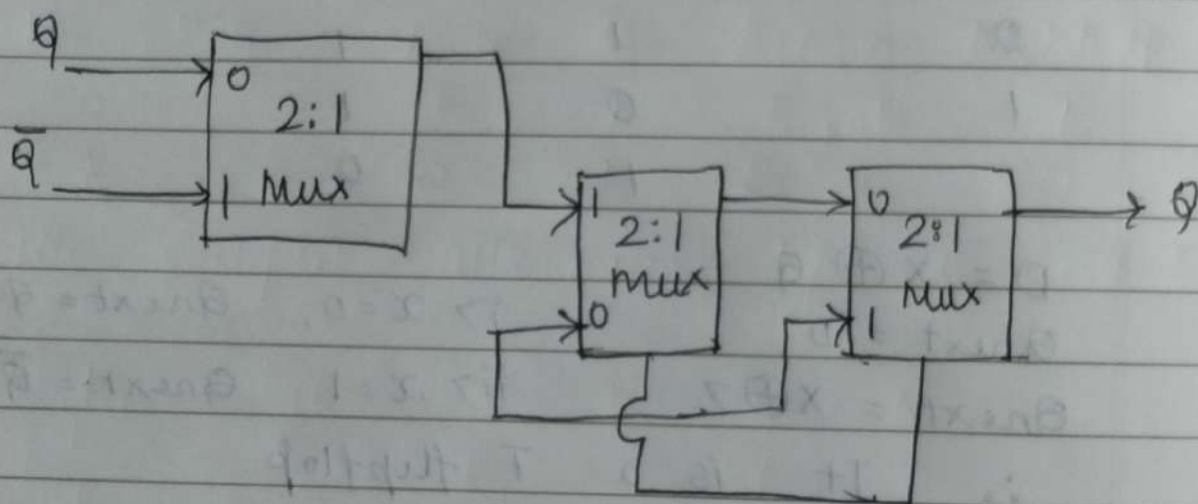
Master enable = $\overline{CLK}$

Slave enable = CLK

$CLK = 0$ master tracks i/p, slave holds

$CLK = 1$ master slaves copies masters output

3. Design a nedge triggered T flipflop using 2:1 Muxes only



4) Convert by adding external gates:

a) A D flipflop to a JK flipflop

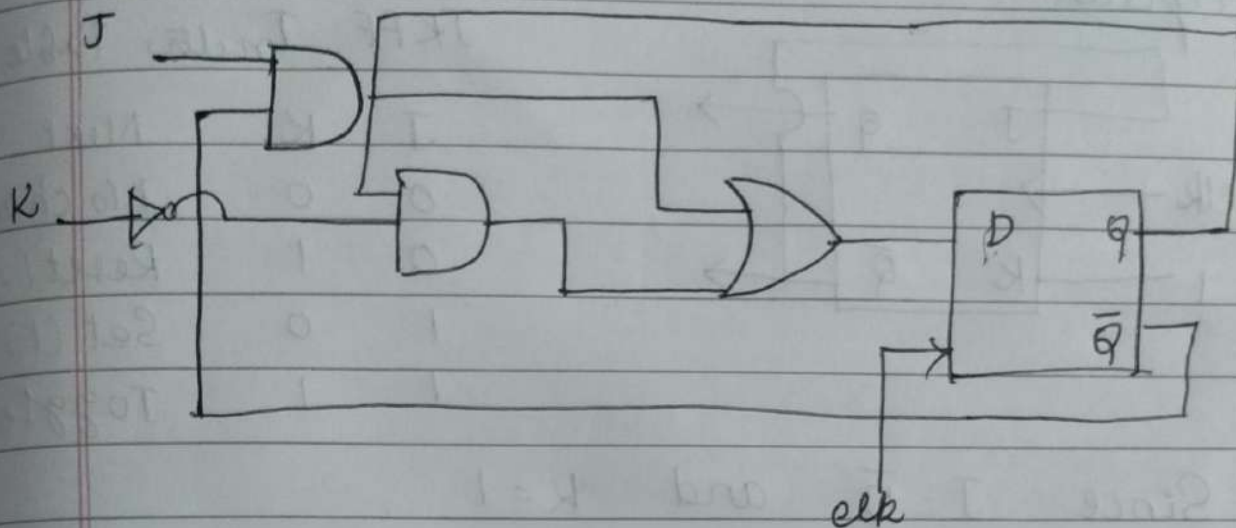
b) A T flipflop to a D flipflop

a) A D flipflop to a JK flipflop
characteristic eqⁿ of JK

$$Q_{n+1} = J\overline{Q} + \overline{K}Q$$

DFF: $D = Q_{n+1}$

$$\Rightarrow D = J\overline{Q} + \overline{K}Q$$



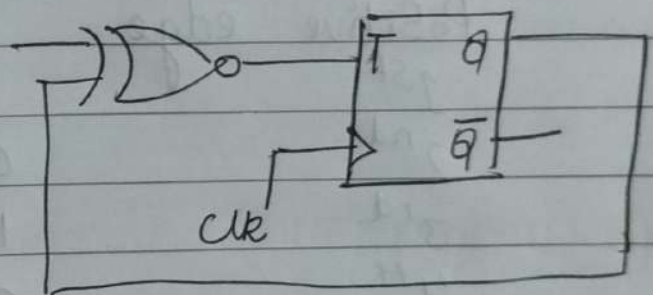
b) A T flipflop to a D flipflop
Characteristic eqⁿ of TFF

$$Q_{n+1} = T \oplus Q$$

$$\text{DFF : } Q_{n+1} = D$$

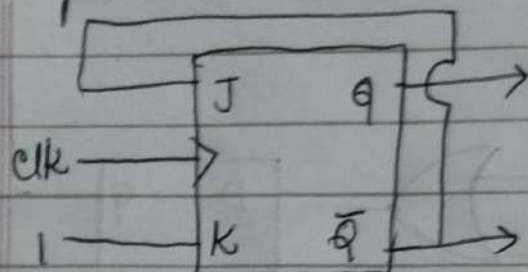
$$\Rightarrow D = T \oplus Q$$

$$T = D \oplus Q$$



5) In a positive edge triggered JKFF, we have $J = Q'$ and $K = 1$ as shown below. Assume the flipflop was initially cleared and then clocked for 6 clock pulses, find the sequence at the Q output for those 6 clock

cycles.



JKFF Truth Table

J	K	Next
0	0	No change
0	1	Reset (0)
1	0	Set (1)
1	1	Toggle

Since $J = \bar{Q}$ and $K = 1$

Present Q	$J = \bar{Q}$	K	Action	Next Q
0	1	1	Toggle	1
1	0	1	Reset	0

The output alternates b/w 0 and 1

Positive edge

	J	K	Q	$\bar{Q}$
1 st	1	1	1	0
2 nd	0	1	0	1
3 rd	1	1	1	0
4 th	0	1	0	1
5 th	1	1	1	0
6 th	0	1	0	1

$\therefore$ The sequence $Q = 1, 0, 1, 0, 1, 0$.